

★ NUMBER SYSTEMS + RATIONALITY ★

Quant | Ch01 Numbers + Divisibility | Topic 1 of 12

► Meaning in one line

Natural (1,2,3..) , whole (, 0) , integers ($\pm$) , rational (p/q).

► Key points

Point	Example
$N \subset W \subset Z \subset Q$; closed + ordered + number-line.	-5 integer, 0 whole, $7/3$ rational.
p/q ($q \neq 0$) + terminating/repeating; else irrational.	0.333.. rational; π irrational.

★ IMPORTANT FOR EXAM

Prime + 1 + composites + co-prime (gcd 1!) + even/odd + parity —

⚠ TRAP - read twice

$\sqrt{4} = 2$ (rational!) + ' $\pi = 22/7$ exactly' + 0-natural + fraction-always-rational confusions! $22/7$ approximates; $\sqrt{9}$ exits surds.

✓ **Try yourself:** Classify: -3, 0, $5/2$, $\sqrt{2}$, 0.4545..? **Answer:** -3 Z; 0 W; $5/2$ Q; $\sqrt{2}$ irrational; 0.4545 Q.

★ PRIMES + PARITY ★

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► Meaning in one line

Prime (2 only-even).

► Key points

Point	Example
prime 2 factors; test to $\sqrt{n}$; 1 neither.	91 = 7×13 composite; (8,15) co-prime.
T2 rules: $E \pm E = E$, $O \pm O = E$, $E \pm O = O$; $E \times \text{any} = E$.	odd+odd+even = even.

★ IMPORTANT FOR EXAM

Place/face + factors/multiples + remainder

⚠ TRAP - read twice

1-prime + 2-composite + '91 prime' + even-prime hunt misses! Test $\sqrt{n}$ + sieve.

✓ **Try yourself:** Prime? 1, 2, 51, 97. Odd/even? $13+17+20$. **Answer:** 2, 97 prime; 1 neither; $51 = 3 \times 17$; sum = even.

★ PLACE VALUE + FACTORS/MULTIPLES ★

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► Meaning in one line

Place (position) ×.

► Key points

Point	Example
value = digit × place; expanded + compare + round.	7 in 876 = 70; face = 7.
a b + times-table + first-common.	factors(12) = 1,2,3,4,6,12; multiples(6) = 6,12,18..

★ IMPORTANT FOR EXAM

Remainder + divisibility + HCF/LCM

⚠ TRAP - read twice

Place ↔ face swap + '0 adds value' + factor/multiple-flip + 1-factor-everything misses! Position + divide.

✓ **Try yourself:** 7 in 5708 (place+face)? Factors of 18? **Answer:** Place 700, face 7; 1,2,3,6,9,18.

★ REMAINDER + DIVISIBILITY BY 2 ★

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► Meaning in one line

Dividend = divisor $\times$ quotient , remainder ($0 \leq r$ less-than-d.

► Key points

Point	Example
$D = dq + r$; verify + remainder-theorem.	$47 \div 6$: q7 r5; check $6 \times 7 + 5 = 47$.
units 0/2/4/6/8 $\rightarrow$ divisible.	3,458 even-yes; 7,731 no.

★ IMPORTANT FOR EXAM

Div-3/4/5/6

⚠ TRAP - read twice

$r \geq d$ -accept + 'remainder 0 = prime' + odd/even-last

✓ **Try yourself:** Divide 100 by 7 (q+r)? Divisible by 2: 4,506? **Answer:** q14 r2 (98+2); 4,506 yes (ends 6).

★ DIVISIBILITY BY 3 + 4 ★

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► Meaning in one line

Div-3: digit-sum $\div 3$.

► Key points

Point	Example
sum digits; if $\div 3$ yes.	4,671 $\rightarrow$ 18 $\rightarrow$ yes; 7,732 $\rightarrow$ 19 $\rightarrow$ no.
last-two $\div 4$.	3,124 (24 yes); 5,678 (78 no).

★ IMPORTANT FOR EXAM

Div-5/6 + combined-rules

⚠ TRAP - read twice

Whole-number $\div 3$ + 'last-digit $\div 3$ ' + div-3/4-swap

✓ **Try yourself:** Div-3: 9,453? Div-4: 8,936? **Answer:** $9+4+5+3 = 21$ yes; $36 \div 4 =$ yes.

★ DIVISIBILITY BY 5 + 6 ★

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► Meaning in one line

Div-5: ends 0/5.

► Key points

Point	Example
units 0/5.	7,890 yes; 4,567 no.
even + digit-sum $\div 3$.	4,572 (even + 18 yes); 3,471 (odd no).

★ IMPORTANT FOR EXAM

Div-8/9 + powers-of-2/3

⚠ TRAP - read twice

Div-6 = $\div 6$ -last + 'either 2-or-3' + no-both

✓ **Try yourself:** Div-5: 12,345? Div-6: 7,794? **Answer:** Ends 5 yes; even + 27 yes → yes.

★ DIVISIBILITY BY 8 + 9 ★

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► Meaning in one line

Div-8: last-three $\div 8$ (halve-thrice).

► Key points

Point	Example
last-three $\div 8$ (or halve $\times 3$).	7,336 ($336 \div 8 = 42$ yes); 5,678 no.
digit-sum $\div 9$.	8,019 $\rightarrow 18$ yes; 4,671 $\rightarrow 18$ — also yes!

★ IMPORTANT FOR EXAM

Div-10/11 + alternating

⚠ TRAP - read twice

Last-two $\div 8$ + 'div-9 = div-3'-miss

✓ **Try yourself:** Div-8: 9,168? Div-9: 33,660? **Answer:** $168 \div 8 = 21$ yes; sum 18 yes.

★ DIVISIBILITY BY 10 + 11 ★

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► Meaning in one line

Div-10: ends 0.

► Key points

Point	Example
units 0.	56,780 yes; 9,876 no.
odd-place sum – even-place sum $\div 11$.	918,082: $(9+8+8)-(1+0+2) = 25-3 = 22$ yes.

★ IMPORTANT FOR EXAM

Missing-digit + least-add + HCF

⚠ TRAP - read twice

Div-11 = $\div 11$ -number + 'digits $\div 11$ ' + place-mix

✓ **Try yourself:** Div-10: 40,905? Div-11: 2728? **Answer:** Ends 5 no; $(2+2)-(7+8) = 4-15 = -11 \rightarrow$ yes.

★ MISSING-DIGIT + LEAST ADD/SUBTRACT ★

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► Meaning in one line

Replace , test-0-9 , rule-fit.

► Key points

Point	Example
* + rule + test digits.	$4*2 \div 3 \rightarrow 4+2 = 6$, need $3/12 \rightarrow *=1,4,7$.
least = next-multiple – N.	least + 476 to $\div 11$: $44 \times 11 = 484 \rightarrow +8$.

★ IMPORTANT FOR EXAM

Prime-factor + division-HCF + product

⚠ TRAP - read twice

First-digit-guess + 'any fits' + no-rule/verify

✓ **Try yourself:** $2*5 \div 11$ (find *)? Least + 1,000 to $\div 9$? **Answer:** *: $(2+5)=7$ need $4/15 \rightarrow *=4$; $1008-1000 = +8$.

★ PRIME FACTORIZATION + DIVISION-HCF ★

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► Meaning in one line

Tree , divide-by-primers , exponents , ladder.

► Key points

Point	Example
tree/ladder + exponents.	$72 = 2^3 \times 3^2$; verify multiply-back.
divide + remainders + HCF-last-divisor.	HCF(48,180): $180 \div 48 \dots 3r36$; $48 \div 36 \dots 1r12$; $36 \div 12 = 3 \rightarrow 12$.

★ IMPORTANT FOR EXAM

Product + HCF/LCM-words

⚠ TRAP - read twice

1-prime + missed-exponents + 'HCF = small'

✓ **Try yourself:** Factor 96? HCF(36, 60) by division? **Answer:** $96 = 2^5 \times 3$; $60 \div 36 \text{ r}24$; $36 \div 24 \text{ r}12$; $24 \div 12 = 2 \rightarrow 12$.

★ PRODUCT RULE + HCF/LCM WORDS ★

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► Meaning in one line

$a \times b = \text{HCF} \times \text{LCM}$, common-divide , ratio-split.

► Key points

Point	Example
$a \times b = H \times L$ (two numbers!).	H6 L72, a18 $\rightarrow b = 6 \times 72 / 18 = 24$.
greatest/divide = HCF; least/together = LCM.	bells 12/18/20 $\rightarrow \text{LCM } 180 \text{ min} = 3 \text{ h}$.

★ IMPORTANT FOR EXAM

Fractions + decimals + BODMAS

⚠ TRAP - read twice

Product-for-3-numbers + HCF/LCM-swap + 'bigger = LCM'

✓ **Try yourself:** H8 L60, a20 $\rightarrow b$? Tiles 15/25 together? **Answer:** $b = 8 \times 60 / 20 = 24$; $\text{LCM}(15, 25) = 75$.

★ FRACTION TYPES (ENTRY) ★

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► Meaning in one line

Proper , improper , mixed , like , unlike , equivalent , lowest , convert , compare.

► Key points

Point	Example
★ IMPORTANT FOR EXAM	
Operations + decimals + percent	
⚠ TRAP - read twice	
Numerator/denominator-flip + 'bigger-denominator = bigger' + no-reduce	

✓ Try yourself: $7/3$ to mixed? Lowest of $12/18$? Answer: $2\frac{1}{3}$; divide 6 $\rightarrow 2/3$.